Duration of Dual Antiplatelet Therapy in Patients With Coronary Artery Disease

Objective:

To review the optimal duration of dual antiplatelet therapy (DAPT) following elective and emergent percutaneous coronary intervention (PCI) for the treatment of chronic stable angina and acute coronary syndromes (ACS).

Background:

The optimal duration of DAPT following PCI is influenced by patient, clinical and procedural characteristics. The benefit of DAPT for reduction in ischemic risk must be weighed against the risk of bleeding from its extended use. This document summarizes recent changes in clinical practice guidelines.

Efficacy Of Dual Antiplatelet Therapy

The use of acetylsalicylic acid (ASA) in combination with either clopidogrel, ticagrelor, or prasugrel is indicated for the treatment of ACS to prevent future major adverse cardiovascular events (MACE). The use of ASA in combination with clopidogrel, ticagrelor or prasugrel is indicated following emergent PCI for the treatment of ACS, and the use of ASA with clopidogrel is indicated following elective PCI for the treatment of chronic stable angina.

DAPT in those with ACS who underwent revascularization

- The **PCI-CURE** study initially established the benefit of DAPT (clopidogrel with ASA) over ASA alone in patients who underwent PCI for Non-ST Elevation Acute Coronary Syndrome (NSTEACS).
- The **PLATO** trial (ticagrelor) and **TRITON-TIMI 38** trial (prasugrel) demonstrated increased efficacy of the second generation P2Y12 inhibitors over clopidogrel following PCI for an ACS indication. Ticagrelor also demonstrated efficacy amongst ACS treated with bypass surgery.
- The **ISAR-REACT 5** trial compared DAPT with ticagrelor vs DAPT with prasugrel in patients who presented with ACS and were undergoing revascularization. The prasugrel regimen demonstrated a 26% relative risk reduction over the ticagrelor regimen in a composite CV outcome, which was driven by nonfatal MI.

Efficacy of DAPT in those with ACS who were medically managed

Clopidogrel used in combination with ASA for the medical treatment of ACS in the CURE trial resulted in a 20% reduction in the combined endpoint of death, myocardial infarction, and stroke. The PLATO trial (ticagrelor) demonstrated increased efficacy of ticagrelor over clopidogrel amongst ACS treated DAPT as part of medical management. Prasugrel did not show additional benefit over clopidogrel when used for the treatment of ACS without revascularization in the TRILOGY trial.

None of these trials showed an overall mortality benefit with DAPT.

Safety Of Dual Antiplatelet Therapy (Bleeding Considerations)

While second generation PG1Y12 inhibitors (prasugrel and ticagrelor) reduce MACE rates cor
in patients receiving ASA in the setting of ACS and PCI, these agents are associated with incre

bleeding. Compared to clopidogrel, ticagrelor was associated with an increased risk of non-CA
in the PLATO trial, and prasugrel was similarly associated with an increased risk of major blee
trial. Use of prasugrel was associated with similar rates of bleeding amongst medically treated
TRILOGY trial. As such, caution needs to be exercised when choosing to use DAPT and when selecting a second generation P2Y12 inhibitor over clopidogrel; especially in patients at increased risk of bleeding. Table 1 lists clinical variables that have been associated with increased bleeding in clinical trials and registries.

Table 1: Major and Minor Variables Associated With An Increased Risk Of Bleeding Post PCI

Duration Of Dual Antiplatelet Therapy Following ACS

DAPT durations less than 12 months

The results of the CURE and PCI-CURE trials led to the recommendation for 1 year of DAPT using clopidogrel and ASA for the treatment of ACS, as an adjunct to medical therapy, CABG or PCI. Amongst ACS patients treated with PCI, numerous trials of selected patients have now illustrated that shorter durations of DAPT (as short as 1 or 3-6 months) post PCI are associated with reductions in clinically relevant and major bleeding compared with 12 months of DAPT. Similarly, de-escalation of potent P2Y12 inhibitor-based (ticagrelor or prasugrel) DAPT at 30 days to clopidogrel-based DAPT results in less major bleeding. The single antiplatelet therapy studied after DAPT has either been ASA or a P2Y12 inhibitor.

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Few studies have demonstrated that shorter durations of DAPT following PCI for an ACS indication have been associated with higher rates of stent thrombosis and recurrent myocardial infarction (MI). However, it's crucial to acknowledge that much of the evidence for these strategies in ACS patients comes from trials primarily focused on bleeding outcomes, many employing a non-inferiority design that may not adequately detect relevant differences in ischemic outcomes. It is also important to consider that the enrolled patient populations in these studies were often selectively chosen, potentially excluding or under-representing the highest-risk ACS patients. As such, it is currently suggested that DAPT be continued for at least 1 year post ACS in patients who are not at high risk for bleeding.

From a practical standpoint, DAPT strategies under 12 months should not be universally adopted in the broader ACS population by default. Instead, they should be considered when there is a specific rationale for their use, such as aiming to reduce bleeding risk in high bleeding risk patients or addressing specific concerns about a 12-month potent P2Y12 inhibitor-based DAPT regimen. Moreover, while de-escalation of antiplatelet therapy within the first 30 days is not recommended, considering de-escalation of P2Y12 receptor inhibitor therapy as an alternative strategy beyond 30 days after ACS may be appropriate to minimize bleeding risk.

Overall, DAPT provides protection against ischemic events. However, individuals experiencing significant bleeding after PCI face a heightened mortality risk that has the potential to counterbalance the protective effects of DAPT, particularly in patients at high bleeding risk. While the current standard remains 12-month DAPT, in those at heightened risk for bleeding or experiencing bleeding events, clinicians should favor a shared-decision decision-making approach weighing the risks of bleeding versus ischemic events with a short duration or de-escalation strategy.

Below is a summary of recent trials that have assessed the safety of de-escalation strategies and short duration DAPT in patients treated with PCI for both stable coronary artery disease and ACS. See **Figure 1**.

- **STOP-DAPT2** investigated 1 month of DAPT, followed by clopidogrel monotherapy vs continued DAPT in patients mostly at low-to-medium thrombotic risk. Superiority was found with the clopidogrel monotherapy group in a large composite outcome of cardiovascular thrombotic and bleeding events. The major secondary endpoints suggested that 1 month of DAPT was non-inferior to 12 months of DAPT in terms of cardiovascular endpoints and superior in terms of bleeding. There was no significant subgroup interaction in patients who experienced ACS as the reason for PCI. However, there was a significant subgroup interaction those with an estimated GFR less than 30 mL/min/1.73m2, suggesting those patients may have better outcomes with 12 months of DAPT rather than 1 month.
- **SMART-CHOICE** studied 3 months of DAPT, followed by P2Y12 monotherapy (clopidogrel used in ~75% of patients) compared with 12 months of DAPT. This study found similar outcomes to STOP-DAPT2, with noninferiority demonstrated for MACE. This was consistent in the subgroup of patients who experienced ACS. There were similar rates of major bleeding between the two groups.
- The much larger **TWILIGHT** study (about 7000 patients) assessed 3 months of DAPT with ticagrelor, followed by ticagrelor monotherapy vs continued DAPT. The trial population had at least 1 feature associated with a high risk of ischemic or bleeding events. There was a significant reduction in bleeding in those assigned to 3 months of DAPT. This study also suggested non-inferiority with 3 months of DAPT in terms of MACE. There was no significant interaction in the subgroup who underwent PCI for ACS.
- **GLOBAL LEADER**S was an open-label trial of 15,968 patients that compared ticagrelor plus aspirin for 1 month, followed by ticagrelor monotherapy for 23 months versus aspirin plus clopidogrel (75 mg daily for stable CAD) or ticagrelor (90 mg twice daily for ACS) for 12 months, followed by aspirin monotherapy for 12 months after implantation of a drug-eluting stent. Ticagrelor monotherapy compared with aspirin reduced the secondary composite outcome of all- cause death, MI, and stroke at the cost of increased BARC 3 or

- In the **STOPDAPT-2 ACS** trial, clopidogrel monotherapy after 1 month of DAPT appeared t
risk of ischemic events (primarily MI) compared with 12 months of DAPT.
- The **MASTER DAPT** trial compared abbreviated DAPT (1 month) followed by either aspirir
monotherapy vs. DAPT ≥3 months (standard therapy) in a cohort of 4579 high bleeding risk patients with ACS undergoing PCI. Overall, adverse clinical outcomes and major adverse cardiac or cerebral events were comparable between the groups, whereas major or clinically relevant non-major bleeding events were significantly reduced in the abbreviated therapy group.
- The **SMART-DATE** trial randomized 2,712 patients with ACS undergoing PCI to either 6-month DAPT or 12-month or longer DAPT. DAPT consisted of a P2Y12 inhibitor and aspirin for both groups. Six-month duration of DAPT was found non-inferior to 12-month or longer DAPT in the composite outcome of all-cause death, myocardial infarction, or stroke at 18 months post-PCI. However, the 6-month DAPT group was associated with a higher rate of myocardial infarction but neither all-cause death nor stroke differed significantly among groups.
- The **TICO** trial evaluated ticagrelor monotherapy vs. ticagrelor plus aspirin for up to 1 year after 3 months of DAPT (ticagrelor and aspirin) in 3056 ACS patients. Overall, adverse clinical outcomes and major bleeding events were significantly reduced with ticagrelor monotherapy, and major adverse cardiac and cerebrovascular events were not significantly different.
- Two randomized control trials, the **TOPIC** and **TALOS-AMI** trials, evaluated de-escalation from a potent P2Y12 inhibitor based DAPT strategy to clopidogrel-based DAPT at 30 days post PCI. The TOPIC trial found no significant difference in ischemic outcomes, but a reduction in major bleeding events with this de-escalation. The TALOS-AMI trial found that de-escalation to clopidogrel-based DAPT significantly reduced the composite of ischemic outcomes or major bleeding. Pooled analysis of these trials illustrates that de-escalation to clopidogrel-based DAPT was associated with similar ischemic outcomes and a significant reduction in major bleeding.
- The **4D-ACS trial** (of 656 East Asian ACS patients) showed that one-month DAPT with ASA and 10 mg prasugrel followed by de-escalation to prasugrel 5 mg monotherapy in ACS patients who received a DES showed a reduction in net adverse clincial events (NACE) by 49%, driven in large part by a 77% reduction in bleeding events compared to the 12-month DAPT regimen. While an ethnically homogenous trial, the principle of efficacy for quick de-escalation of DAPT by dose reduction of the P2Y12 inhibitor was supported.
- The **2025 REC-CAGEFREE II** multicentre, randomised, open label, assessor blind trial indicated the non-inferiority of stepwise dual antiplatelet therapy de-escalation in patients after drug coated balloon angioplasty, suggesting the benefits of de-escalation therapy can be extended to drug coated balloon angioplasty as well.

Most recently, the 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients with Acute Coronary Syndromes (ACS) does recognize the need for a risk-adapted, dynamic approach to DAPT post ACS and incorporates it into there new guideline recommendations.

Figure 1: Alternative antiplatelet strategies in the first 12 months after an ACS

Redrawn from Bainey KR, et al. Can J Cardiol 2024;40:160-181.

Prolonging antithrombotic therapy beyond 12 months (Figure 4)

At 12-months the default approach has been to de-escalate antithrombotic therapy to single antiplatelet agent with ASA. Summarized below are the alternative strategies for antithrombotic therapy beyond 12-months in patients with prior CAD including those with prior ACS and PCI. This includes extended duration DAPT and dual pathway inhibition with the use of low-dose anticoagulation.

Figure 2: Antithrombotic therapy beyond the first 12 months after ACS

Adapted from Byrne RA, et al. Eur Heart J 2023;44:3720-3826.

DAPT durations more than 12 months

Increasing evidence has suggested that patients at high risk for MACE following PCI for ACS may derive benefit from a strategy of extended DAPT beyond 1 year. The **DAPT** trial evaluated a strategy of 30 vs 12 months of DAPT (clopidogrel or prasugrel in combination with ASA) following PCI for either stable CAD or ACS. Compared to 12 months of DAPT, 30 months of DAPT was associated with lower rates of stent thrombosis and MACE. However, extended use of DAPT was associated with higher rates of moderate to severe bleeding. No cardiovascular-related mortality difference was found. There was an unexpected exploratory finding of higher rates of all-cause mortality associated with extended DAPT.

Similar results were obtained in the **PEGASUS** trial, which enrolled stable patients who had experienced an MI within 1-3 years to one of two doses of ticagrelor (either 90mg BID or 60mg BID) or placebo in addition to ASA. The median duration of follow up was 33 months, and the median time on ticagrelor was 29 months. The composite rate of cardiovascular death, myocardial infarction and stroke was reduced by both doses of ticagrelor compared to placebo. However, both doses of ticagrelor increased the risk of TIMI major bleeding compared to placebo. There was no difference in mortality between the groups. **Table 2** lists clinical variables that may help to identify those patients in whom the potential for bleeding may outweigh any efficacy gains with extended DAPT.

In summary, extended DAPT for more than 12 months decreases thrombotic outcomes at the expense of a higher risk of bleeding. There is no clear mortality benefit. The decision to extend DAPT beyond 12 months requires individualization based on thrombotic and bleeding risk factors. **Figure 3** outlines the current Canadian Cardiovascular Society recommendations for DAPT for ACS patients who undergo PCI.

Dual Pathway Inhibition with Low-Dose Anticoagulation After DAPT

The **COMPASS** trial, involving 27,395 stable CAD patients (62% with a history of MI), suggests considering a dual antithrombotic therapy strategy for maintenance treatment beyond 12 months post-ACS. This strategy, comprising a very low dose of rivaroxaban (2.5 mg b.i.d.) with aspirin, is recommended for patients at high thrombotic risk without a significant bleeding risk increase, and may be considered for those with moderately elevated thrombotic risk. The rivaroxaban 2.5 mg b.i.d. plus aspirin combination showed a significant reduction in combined ischemic events, overall mortality, and cardiovascular mortality. However, it increased major bleeding wi
rise in fatal, intracranial, or critical organ bleeding events. [see **Clinical Guide:** Rivaroxaban (

COMPASS-PCI a subgroup analysis of COMPASS examined the outcomes of patients with a syndrome with or without a previous PCI. Amongst patients with prior PCI, dual pathway inhibi aspirin produced significant reductions in MACE irrespective of time since previous PCI (as ea far as 10 years).

Duration Of Dual Antiplatelet Therapy Following Elective PCI For A Non-ACS Indication

Patients who undergo elective PCI for a non-ACS indication are at inherently lower risk for long-term MACE compared to those who undergo PCI for an ACS indication. In addition, newer generation drug eluting stents (DES) have inherently lower thrombotic risk compared to older first generation sirolimus and tacrolimus DES. Several trials have demonstrated the safety of a shorter course of DAPT in this setting, although increased thrombotic risk has been associated with several angiographic and clinical variables (**Table 2**). As such, although DAPT is recommended for up to 1 year post elective PCI, its use may also be shortened to a total of 6 months. DAPT use beyond 1 year may be considered if a patient has additional clinical or angiographic variables that suggest an increased risk of a thrombotic event and is a low risk for bleeding (**Tables 1 and 2**).

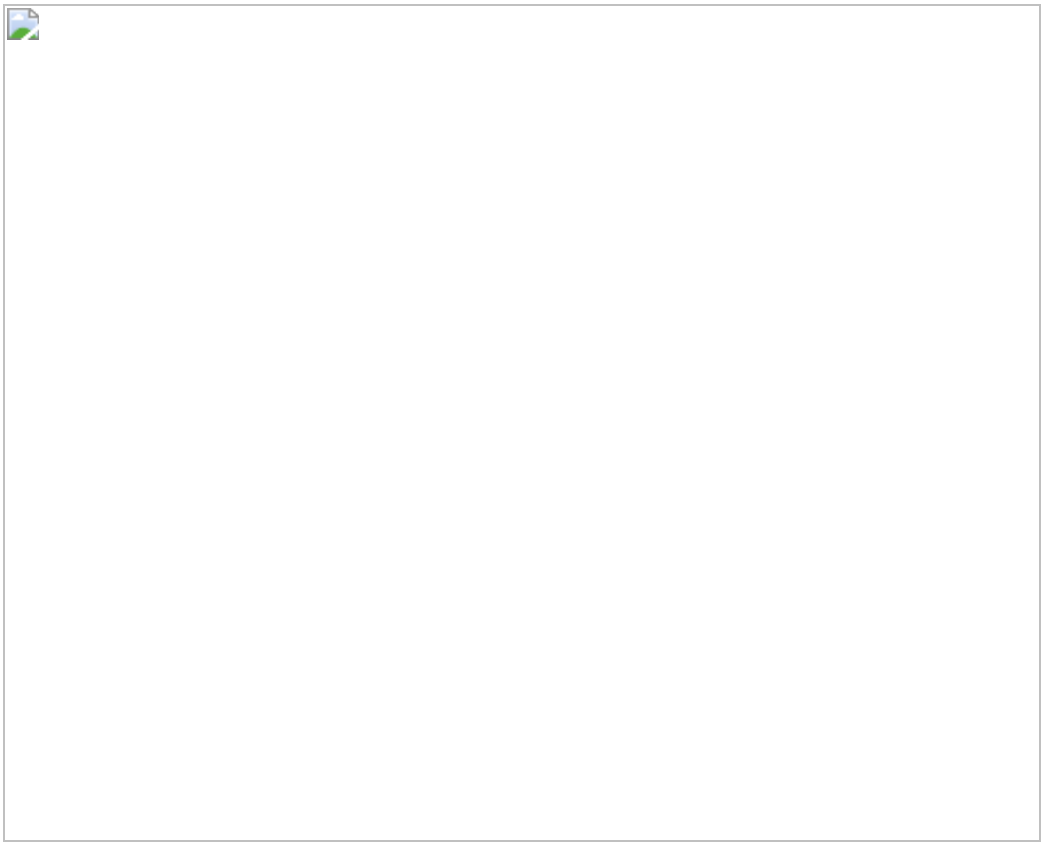
Table 2: Variables Associated With A Higher Risk Of A Thrombotic Event Post PCI For ACS

Duration Of DAPT In Patients with an Indication for Anticoagulation Post ACS

Patients often have an indication for anticoagulation post ACS, such as atrial fibrillation, VTE or mechanical valve. Meta-analyses across randomized trials suggest no difference in mortality, stroke, and overall MACE when aspirin is discontinued for patients who are on an oral anticoagulant, although there is a marginal apparent increase in MI and stent thrombosis. The 2025 ACC Guidelines reaffirm a 1B recommendation for discontinuing ASA within 1-4 weeks of beginning triple antithrombotic therapy. Trials of the more potent second generation P2Y12 inhibitors excluded such patients, and therefor a switch to clopidogrel is commonly used as the P2Y12 inhibitor in this scenario.

Figure 3 outlines the current Canadian Cardiovascular Society recommendations for DAPT therapy for patients with ACS. **Figure 4** outlines the 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline recommendations for DAPT Therapy for patients with ACS.

Figure 3: CCS Current Recommendations For Duration Of Dapt In Patie (STEMI Or NSTEACS) Who Undergo PCI



[1] Factors associated with increased bleeding risk include: need for OAC in addition to DAPT, advanced age (>75 years), frailty, anemia with hemoglobin <110 g/d, chronic renal failure (creatinine clearance <40 mL/min), low body weight (<60 kg), hospitalization for bleeding within last year, prior stroke/intracranial bleeding, regular need for NDAISs or prednisone.

[2] Instead of ticagrelor or clopidogrel, prasugrel 5-10 mg daily is also an option (weak recommendation).

BID, twice daily; DAPT, dual antiplatelet therapy; NSTEACS, non-ST segment elevation acute coronary syndrome; NSTEMI, non-ST elevation myocardial infarction; OD, once daily; SAPT, single antiplatelet therapy; STEMI, ST segment elevation myocardial infarction

Figure 4: 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline Recommendations For Duration Of DAPT In Patients With ACS (STEMI Or NSTEACS) Who Undergo PCI

Green =1A Recommendation; Orange = 2B Recommendation

Redrawn from Rao SV, et al. 2025 ACC/AHA/ACEP/NAEMSP/SCAI guideline for the management of patients with acute coronary syndromes: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. J Am Coll Cardiol. 2025;85(22):2135-2237.

Figure 5: Recommendations For Duration Of DAPT In Patients Undergoing Elective PCI

[1] Factors associated with increased bleeding risk include: need for OAC in addition to DAPT, advanced age (>75 years), frailty, anemia with hemoglobin <110 g/d, chronic renal failure (creatinine clearance <40 mL/min), low body weight (<60 kg), hospitalization for bleeding within last year, prior stroke/intracranial bleeding, regular need for NDAISs or prednisone.

[2] Clinical and angiographic features associated with increased risk of thrombotic events include: age >65, diabetes mellitus, prior myocardial infarction, chronic renal dysfunction (creatinine clearance <60 mL/min), multi-vessel disease, multiple stents implanted, complex bifurcation lesions, total stent length >60 mm, chronic total occlusion intervention or bioabsorbable vascular scaffold (BVS) implantation.

BMS, bare metal stent; DAPT, dual antiplatelet therapy; DES, drug-eluting stent; SAPT, single antiplatelet therapy

Other Relevant

- Acetylsalicylic Acid (ASA®)
- Clopidogrel (Plavix®)
- Prasugrel (Effient®)
- Rivaroxaban (Xarelto®)
- Ticagrelor (Brilinta®)

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